Unit Test Report

Automated API/unit tests for Sprint 1 and Sprint 2 (US01–US21), run against a seeded database.

1. Summary

Total tests	Passed	Failed	Skipped	Suites	Duration
57	57	0	0	14	8.2 s

2. Test environment

Item	Value
Project	Online Exam Portal (github.com/satvikkalyanpur/Online-Exam-Portal), commit f677021
Date of execution	06 November 2026
Runtime	Node.js v22.23.4, tsx test runner (node:test), Prisma 7.10.1
Database	MariaDB 10.11.14 (the project documents MySQL 8; the app uses the MariaDB driver adapter, which supports both)
Test data	Seeded with src/seed.ts (4 exams, 12 questions, 2 finished attempts, 4 accounts)
Test command	npm test (tsx --test --test-concurrency=1 "test/*.test.ts")

Environment note: the Prisma schema-engine binary could not be downloaded in the execution sandbox, so the database tables were created from database/schema.prisma using equivalent hand-written DDL instead of 'prisma db push'. The Prisma client was generated normally and the seed script ran unchanged.

3. Results by suite

Suite	Test case	Result	Time (ms)
US01 login	valid credentials log the user in and return the role	Pass	120.1
	a wrong password and an unknown email get the same message	Pass	132.6
	missing fields are rejected	Pass	3.8
US02 role-based access	the role is identified after login	Pass	17.3
	requests without a valid token are rejected	Pass	6.3
	a student cannot use teacher or administrator features	Pass	37.9
	a teacher cannot create exams, see reports or take exams	Pass	23.6
	an administrator cannot edit the question bank or take exams	Pass	25.5
US04 create examination	an administrator creates an exam with title, subject, duration, marks and schedule	Pass	36.5
	invalid details are rejected	Pass	17.9
US05–US07 question bank	US05: a teacher adds questions, linked to an exam or kept for later	Pass	118.5
	US05: invalid questions are rejected	Pass	35.0
	US06: a teacher edits a question and the change is saved	Pass	51.5
	US07: a teacher deletes a question and it is gone	Pass	44.1
US03 view available exams	a student sees published exams with subject, schedule, duration and question count	Pass	14.5
	draft exams are hidden, and moving an exam back to draft hides it	Pass	46.4
US08–US10 taking an exam	US08/US09: starting returns the questions (without answers) and a deadline	Pass	40.6
	US10: an answer is saved when selected, and updated when changed	Pass	121.2
	US08/US10: opening the exam again resumes with the same deadline and answers	Pass	22.6
	a student cannot touch another student's attempt	Pass	11.1
	invalid answers are rejected	Pass	11.1

Suite	Test case	Result	Time (ms)
	US08: the student submits the exam and cannot start it again	Pass	63.7
	an exam with no questions cannot be started	Pass	22.5
US12 restrict access before the scheduled time	an exam that has not opened is listed but cannot be started	Pass	110.7
	the exam can be started once its start time is reached	Pass	95.8
	an exam whose window has ended cannot be started	Pass	53.3
US11, US14 and US15: answering, marking and results	US11: answers to several questions are kept while moving between them	Pass	112.1
	US15: there is no result before the exam is submitted	Pass	9.3
	US14: submitting marks the answers and stores the score	Pass	78.6
	US15: the student sees the result for the right exam	Pass	35.9
	US15/US18: correct answers stay hidden until the exam closes, and results are private	Pass	54.8
	US07: a question that has been marked cannot be deleted	Pass	25.0
	US06: a marked question's wording can be fixed, but not its answer or marks	Pass	43.9
	US16: the administrator sees the attempt in the performance report, filtered by exam	Pass	15.6
	US17: statistics are worked out from the stored results	Pass	10.1
	US16/US17: only administrators can see reports and statistics	Pass	22.9
US13 automatic submission when time runs out	the browser asking for an automatic submission is recorded as automatic	Pass	59.3
	an attempt whose time ran out is submitted even if the student closed the browser	Pass	152.6
	answers sent after the deadline are refused and the attempt is closed	Pass	60.7
	an attempt never runs past the end of the exam window	Pass	47.9
exams that are being taken	an exam's questions cannot change while a student is taking it	Pass	53.9
	a student in the middle of an exam can carry on if it is moved back to draft	Pass	71.6
	answers to questions no longer in the exam are not counted	Pass	96.8
US19 consistency after a lost connection	resending answers and submissions never creates duplicates	Pass	129.8
	an answer racing with the submission is either marked or refused, never stored unmarked	Pass	640.4
	an answer arriving a few seconds after the deadline (network delay) still counts	Pass	91.8
US18 protecting credentials and data	responses carry security headers and are not cached	Pass	5.1
	passwords are stored as bcrypt hashes	Pass	3.1
	repeated wrong passwords are locked out for a while	Pass	26.6
	a refused request is written to the activity log	Pass	9.1
US21 activity logs	logins, exam starts, submissions and exam creation are logged	Pass	18.8
	logs can be filtered by action and searched	Pass	12.7
	only administrators can read the logs	Pass	10.8
US20 backup and recovery	a backup contains the exam and result data	Pass	37.9
	restoring brings back deleted exams, questions and results exactly	Pass	185.5
	a damaged or inconsistent file is refused and nothing changes	Pass	199.8
	only administrators can back up or restore	Pass	24.9

4. Observations

All 57 tests passed with no failures, skips or cancellations. The backup-restore test 'a damaged or inconsistent file is refused and nothing changes' prints a Prisma P2000 (value too long) stack trace to the console; this is the expected rejection path being exercised and the test passes because the restore is rolled back. The test files are sprint1.test.ts (US01–US08, US11 area) and sprint2.test.ts (US11–US21), plus helpers.ts.